

```

... 1
..... name: "Edith Bates",
..... birthday: "07/28/90",
..... address: "7 E. 20th Pl",
..... city: "Chandler",
..... state: "Arizona",
..... position: { name: "Customer Support", remote: false, "full time": true }
..... },
..... {
..... name: "Gwen Harding",
..... birthday: "10/11/86",
..... address: "234 W. 48th St",
..... city: "Phoenix",
..... state: "Arizona",
..... position: { name: "Office Assistant", remote: false, "full time": true }
..... }
... ]});
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId("62e7faab3c33b5dc3815b8cb"),
    '1': ObjectId("62e7faab3c33b5dc3815b8cc"),
    '2': ObjectId("62e7faab3c33b5dc3815b8cd"),
    '3': ObjectId("62e7faab3c33b5dc3815b8ce"),
    '4': ObjectId("62e7faab3c33b5dc3815b8cf"),
    '5': ObjectId("62e7faab3c33b5dc3815b8d0"),
    '6': ObjectId("62e7faab3c33b5dc3815b8d1"),
    '7': ObjectId("62e7faab3c33b5dc3815b8d2"),
    '8': ObjectId("62e7faab3c33b5dc3815b8d3"),
    '9': ObjectId("62e7faab3c33b5dc3815b8d4")
  }
}
Atlas atlas-dgyanu-shard-0 [primary] myFirstDatabase>

```

Alison Davidson now lives at 777 E. 1st St. # 120 Tempe, AZ and Mary Crawley now lives at 8322 W. Vista Pl. Scottsdale, AZ.

```

Atlas atlas-dgyanu-shard-0 [primary] myFirstDatabase> db.employees.updateOne({"_id": ObjectId("62e805563c33b5dc3815b8df")}, { $set : { "address" : "777 E. 1 st # 120" }, {"state": "Az"}, {"city": "Tempe"}, { upsert : true });
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-dgyanu-shard-0 [primary] myFirstDatabase> db.employees.find();
[
  {
    _id: ObjectId("62e805563c33b5dc3815b8df"),
    name: 'Alison Davidson',
    birthday: '04/05/75',
    address: '777 E. 1 st # 120',
    city: 'Gary',
    state: 'Indiana',
    position: { name: 'Customer Support', remote: true, 'full time': true }
  },
  {
    _id: ObjectId("62e805563c33b5dc3815b8e0"),

```

```
Atlas atlas-dgyanu-shard-0 [primary] myFirstDatabase> db.employees.updateOne({"_id": ObjectId("62e805563c33b5dc3815b8e4")}, { $set : { "address" : "8322 W. Vista Pl."}}, {"state": "Az"}, {"city": "Scottsdale"}, { upsert : true});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Atlas atlas-dgyanu-shard-0 [primary] myFirstDatabase>
```

```
{
  _id: ObjectId("62e805563c33b5dc3815b8e4"),
  name: 'Mary Crawley',
  birthday: '07/06/80',
  address: '8322 W. Vista Pl.',
  city: 'Charlotte',
  state: 'North Carolina',
  position: { name: 'Human Resources', remote: true, 'full time': true }
},
{
```

Since all employees now all live in Arizona, there is no need to have a field named "state" within this collection, so please remove it.

```
Atlas atlas-dgyanu-shard-0 [primary] myFirstDatabase> db.employees.updateMany({}, {$unset:{state:""}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 10,
  modifiedCount: 9,
  upsertedCount: 0
}
Atlas atlas-dgyanu-shard-0 [primary] myFirstDatabase> db.employees.find({});
[
  {
    _id: ObjectId("62e805563c33b5dc3815b8df"),
    name: 'Alison Davidson',
    birthday: '04/05/75',
    address: '777 E. 1 st # 120',
    city: 'Gary',
    position: { name: 'Customer Support', remote: true, 'full time': true }
  },
  {
    _id: ObjectId("62e805563c33b5dc3815b8e0"),
    name: 'Henry Chapelton',
    birthday: '09/29/80',
    address: '9324 E. Vista Way',
    city: 'Tempe',
    position: { name: 'Customer Support', remote: false, 'full time': true }
  },
  {
    _id: ObjectId("62e805563c33b5dc3815b8e1"),
    name: 'Alex Miller',
    birthday: '11/22/83',
    address: '244 Price Road',
    city: 'Mesa',
    position: { name: 'Customer Support', remote: false, 'full time': false }
  },
  {
    _id: ObjectId("62e805563c33b5dc3815b8e2"),
    name: 'Carly Nielson',
    birthday: '08/04/87',
    address: '678 W. Westward Road',
    city: 'Phoenix',
    position: { name: 'Office Manager', remote: false, 'full time': true }
  },
  {
    _id: ObjectId("62e805563c33b5dc3815b8e3"),
    name: 'Tom Talbot',
    birthday: '12/30/89',
    address: '12 Oakland Way',
```

Lastly, they would like very efficient searching using the "position" field (remember that field includes a document with three other fields).

```
Atlas atlas-dgyanu-shard-0 [primary] myFirstDatabase> db.employees.find({}, {'position.name': 1, 'position.remote': 1, 'position.full time': 1});
[
  {
    _id: ObjectId("62e805563c33b5dc3815b8df"),
    position: { name: 'Customer Support', remote: true, 'full time': true }
  },
  {
    _id: ObjectId("62e805563c33b5dc3815b8e0"),
    position: { name: 'Customer Support', remote: false, 'full time': true }
  },
  {
    _id: ObjectId("62e805563c33b5dc3815b8e1"),
    position: { name: 'Customer Support', remote: false, 'full time': false }
  },
  {
    _id: ObjectId("62e805563c33b5dc3815b8e2"),
    position: { name: 'Office Manager', remote: false, 'full time': true }
  },
  {
    _id: ObjectId("62e805563c33b5dc3815b8e3"),
    position: { name: 'Inventory Manager', remote: false, 'full time': true }
  },
  {
    _id: ObjectId("62e805563c33b5dc3815b8e4"),
    position: { name: 'Human Resources', remote: true, 'full time': true }
  },
  {
    _id: ObjectId("62e805563c33b5dc3815b8e5"),
    position: { name: 'CEO', remote: false, 'full time': true }
  },
  {
    _id: ObjectId("62e805563c33b5dc3815b8e6"),
    position: { name: 'Intern', remote: false, 'full time': false }
  },
  {
    _id: ObjectId("62e805563c33b5dc3815b8e7"),
    position: { name: 'Customer Support', remote: false, 'full time': true }
  },
  {
    _id: ObjectId("62e805563c33b5dc3815b8e8"),
    position: { name: 'Office Assistant', remote: false, 'full time': true }
  }
]
Atlas atlas-dgyanu-shard-0 [primary] myFirstDatabase> █
```

PART 2

```

Atlas atlas-dgyanu-shard-0 [primary] myFirstDatabase> db.songs.insertMany([ { songId: 1, title: "Girls Just Want To
Have Fun", details: { artist: "Cyndi Lauper", album: "She's So Unusual", releaseYear: 1983 } }, { songId: 2, titl
e: "Hips Don't Lie", details: { artist: "Shakira feat. Wyclef Jean", album: "Oral Fixation Vol. 2", releaseYear: 2
006}}, {songId:3, title:"Poker Face", details:{artist:"Lady Gaga", album:"The Fame", releaseYear: 2008}}, {songId:4
, title:"Wannabe", details:{artist: "Spice Girls", album:"Spice", releaseYear: 1996}}, {songId:5, title:"California
Gurls", details:{artist:"Katy Perry feat. Snoop Dogg", album:"Teenage Dream", releaseYear:2010}}, {songId:6, title
:"Bye, Bye, Bye", details:{artist:"NSYNC", album:"No Strings Attached", releaseYear:2000}}, {songId:7, title:"I Wi
ll Always Love You", details:{artist:"Whitney Houston", album:"I Will Always Love You: The Best of Whitney Houston
", releaseYear:2012}}, {songId:8, title:"Baby One More Time", details:{artist:"Britney Spears", album:"Baby One Mor
e Time", releaseYear:1999}}, {songId:9, title:"Vogue", details:{artist:"Madonna", album:"I'm Breathless", releaseY
ear:1990}}, {songId:10, title:"Rolling in the Deep", details:{artist:"Adele", album:"21", releaseYear:2011}}, {song
Id:11, title:"1234", details:{artist:"Feist", album:"The Reminder", releaseYear:2007}}, {songId:12, title:"Elastic
Heart", details:{artist:"Sia", album:"The Hunger Games: Catching Fire Soundtrack", releaseYear:2015}}, {songId:13,
title:"Oops! I Did It Again", details:{artist:"Britney Spears", album:"Oops! I Did It Again", releaseYear:2000}},
{songId:14, title:"Bad Romance", details:{artist:"Lady Gaga", album:"The Fame Monster", releaseYear:2009}},
... {songId:15, title:"Lose Control", details:{artist:"Missy Elliot", album:"The Cookbook", releaseYear:2005}}, {s
ongId:16, title:"U Can't Touch This", details:{artist:"MC Hammer", album:"Please Hammer, Don't Hurt 'Em", releaseY
ear:1990
..... }}, {songId:17, title:"Thriller", details:{artist:"Michael Jackson", album:"Thriller", releaseYear:1982}}, {s
ongId:18, title:"Single Ladies", details:{artist:"Beyonce", album:"I am... Sasha Fierce", releaseYear:2008}}, {song
Id:19, title:"Rhythm Nation", details:{artist:"Janet Jackson", album:"Janet Jackson's Rhythm Nation 1814", release
Year:1989}}]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId("62e84a073c33b5dc3815b8ef"),
    '1': ObjectId("62e84a073c33b5dc3815b8f0"),
    '2': ObjectId("62e84a073c33b5dc3815b8f1"),
    '3': ObjectId("62e84a073c33b5dc3815b8f2"),
    '4': ObjectId("62e84a073c33b5dc3815b8f3"),
    '5': ObjectId("62e84a073c33b5dc3815b8f4"),
    '6': ObjectId("62e84a073c33b5dc3815b8f5"),
    '7': ObjectId("62e84a073c33b5dc3815b8f6"),
    '8': ObjectId("62e84a073c33b5dc3815b8f7"),
    '9': ObjectId("62e84a073c33b5dc3815b8f8"),
    '10': ObjectId("62e84a073c33b5dc3815b8f9"),
    '11': ObjectId("62e84a073c33b5dc3815b8fa"),
    '12': ObjectId("62e84a073c33b5dc3815b8fb"),
    '13': ObjectId("62e84a073c33b5dc3815b8fc"),
    '14': ObjectId("62e84a073c33b5dc3815b8fd"),
    '15': ObjectId("62e84a073c33b5dc3815b8fe"),
    '16': ObjectId("62e84a073c33b5dc3815b8ff"),
    '17': ObjectId("62e84a073c33b5dc3815b900"),
    '18': ObjectId("62e84a073c33b5dc3815b901")
  }
}

```

- The `title` field needs to be renamed to `songTitle`, so it is clearer to the developers working with the data.

```

Atlas atlas-dgyanu-shard-0 [primary] myFirstDatabase> db.songs.updateMany({}, {$rename:{"title":"songTitle"}});
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 19,
  modifiedCount: 19,
  upsertedCount: 0
}
Atlas atlas-dgyanu-shard-0 [primary] myFirstDatabase> db.songs.find();
[
  {
    _id: ObjectId("62e84a073c33b5dc3815b8ef"),
    songId: 1,
    details: {
      artist: 'Cyndi Lauper',
      album: "She's So Unusual",
      releaseYear: 1983
    },
    songTitle: 'Girls Just Want To Have Fun'
  },
  {
    _id: ObjectId("62e84a073c33b5dc3815b8f0"),
    songId: 2,
    details: {
      artist: 'Shakira feat. Wyclef Jean',
      album: 'Oral Fixation Vol. 2',
      releaseYear: 2006
    },
    songTitle: "Hips Don't Lie"
  },
  {
    _id: ObjectId("62e84a073c33b5dc3815b8f1"),
    songId: 3,
    details: { artist: 'Lady Gaga', album: 'The Fame', releaseYear: 2008 },
    songTitle: 'Poker Face'
  },
  {
    _id: ObjectId("62e84a073c33b5dc3815b8f2"),
    songId: 4,
    details: { artist: 'Spice Girls', album: 'Spice', releaseYear: 1996 },
    songTitle: 'Wannabe'
  },
  {
    _id: ObjectId("62e84a073c33b5dc3815b8f3"),
    songId: 5,
    details: {
      artist: 'Katy Perry feat. Snoop Dogg',
      album: 'Teenage Dream',
      releaseYear: 2010
    },
    songTitle: 'California Gurls'
  },
  {
    id: ObjectId("62e84a073c33b5dc3815b8f4").

```

- They would like to have the `artist` field to be outside the `details` document but the `album` and `releaseYear` should stay within that document.

```
Atlas atlas-dgyanu-shard-0 [primary] myFirstDatabase> db.songs.updateMany({}, {$rename:{"details.artist":"artist"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 19,
  modifiedCount: 19,
  upsertedCount: 0
}
Atlas atlas-dgyanu-shard-0 [primary] myFirstDatabase> db.songs.find();
[
  {
    _id: ObjectId("62e84a073c33b5dc3815b8ef"),
    songId: 1,
    details: { album: "She's So Unusual", releaseYear: 1983 },
    songTitle: 'Girls Just Want To Have Fun',
    artist: 'Cyndi Lauper'
  },
  {
    _id: ObjectId("62e84a073c33b5dc3815b8f0"),
    songId: 2,
    details: { album: 'Oral Fixation Vol. 2', releaseYear: 2006 },
    songTitle: "Hips Don't Lie",
    artist: 'Shakira feat. Wyclef Jean'
  },
  {
    _id: ObjectId("62e84a073c33b5dc3815b8f1"),
    songId: 3,
    details: { album: 'The Fame', releaseYear: 2008 },
    songTitle: 'Poker Face',
    artist: 'Lady Gaga'
  },
  {
    _id: ObjectId("62e84a073c33b5dc3815b8f2"),
    songId: 4,
    details: { album: 'Spice', releaseYear: 1996 },
    songTitle: 'Wannabe',
    artist: 'Spice Girls'
  },
  {
    _id: ObjectId("62e84a073c33b5dc3815b8f3"),
    songId: 5,
    details: { album: 'Teenage Dream', releaseYear: 2010 },
    songTitle: 'California Gurls',
    artist: 'Katy Perry feat. Snoop Dogg'
  },
  {
    _id: ObjectId("62e84a073c33b5dc3815b8f4"),
    songId: 6,
    details: { album: 'No Strings Attached', releaseYear: 2000 },
    songTitle: 'Bye, Bye, Bye',
    artist: 'NSYNC'
  },
  {
    _id: ObjectId("62e84a073c33b5dc3815b8f5"),
    songId: 7,
```